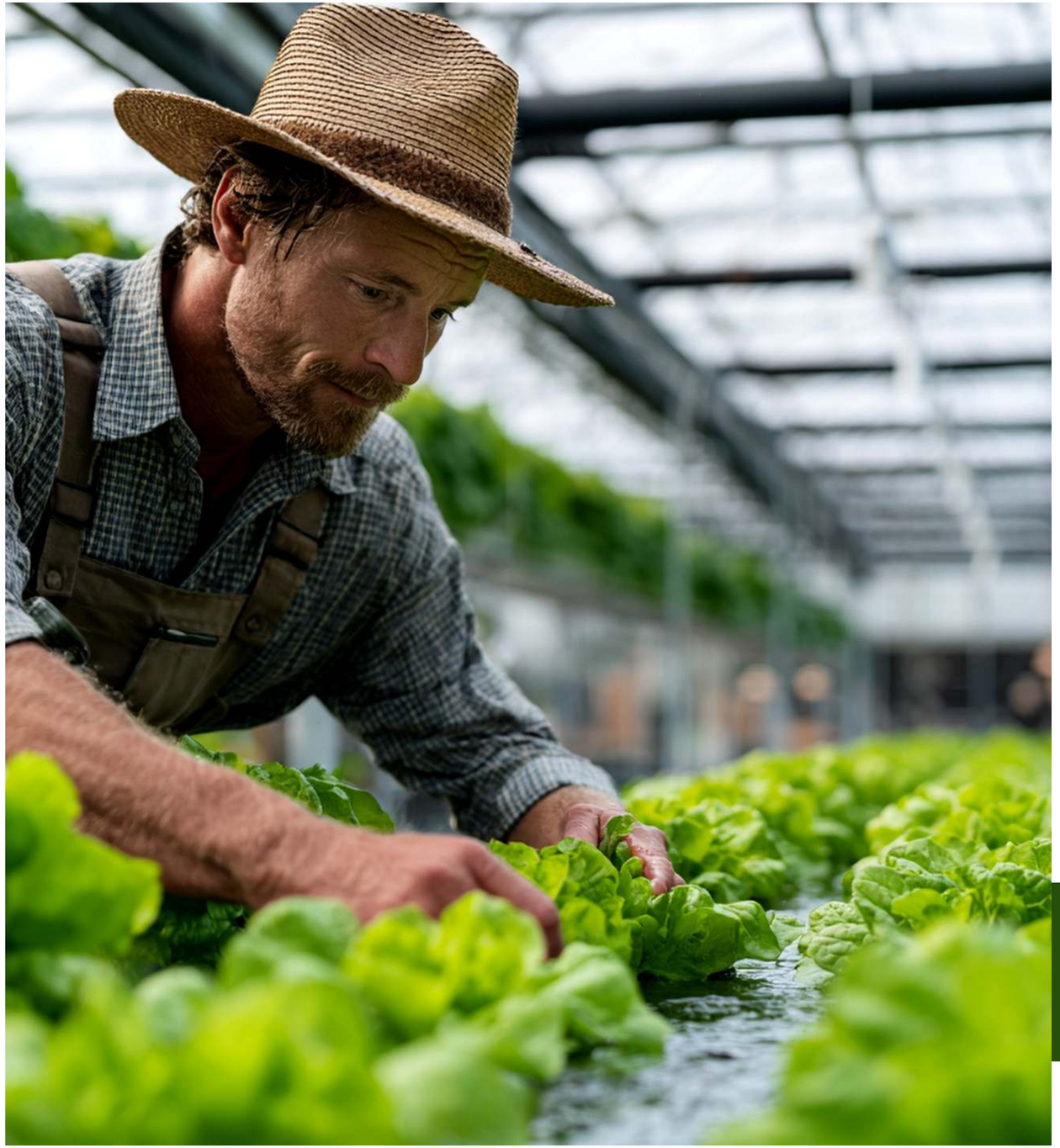
AgriPredict

Innovating for a Sustainable Future

AI





Project Background

- Python-based prototype using machine learning for maize yield estimation
- Built on 7,258 Kenyan field observations (2016–2020)
- Inputs: location, elevation, soil, hybrid seed status
- Dashboard for yield estimates, annual patterns, calculator, and chatbot
- Supports agricultural planning for farmers and advisers

Problem Definition

- Farmers must plan harvest resources without knowing exact yields
- Yield affected by soil, weather, and practices
- Analyzing records is time-consuming and technical
- AgriPredict AI offers experimental yield estimates and planning tools







Project Objectives

- Develop and evaluate a working AI prototype
- Present clear yield estimates to support planning
- Define user needs and system requirements
- Prepare and validate dataset
- Visualize annual yield patterns
- Build and evaluate ML model
- Present results via dashboard and chatbot



Business Success Criteria

- Accepts and validates complete field inputs
- Displays yield estimates and production scenarios
- Provides annual chart comparison
- Makes model error data accessible
- Answers supported questions and explains limitations








Sustainable
Farming Practices







Technology Integration

Project Scope

Included:

- Maize yield prediction via dashboard
- Input validation, model evaluation, visualizations, chatbot

Excluded:

- Other crops, live weather/crop-price forecasts, drone/sensor integration, automated farm control, financial guarantees
-

Tools & Techniques

- Python, pandas, NumPy, Matplotlib, Streamlit, Joblib, Scikit-learn, Canva
- Data cleaning, validation, regression modeling, dashboard design









Expected Business Benefits

- Improved harvest planning
- Clearer production scenarios
- Easy access to historical field info
- Transparent model evaluation
- Accessible explanations via chatbot assistant

Thank You